Adaptive Optimal Control via Q-Learning for Multi-Agent Pursuit-Evasion Games

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***Abstract*—The multi-agent pursuit-evasion (PE) games are solved in this brief to find the best possible strategic options for every participant. In these games, numerous pursuers aim to catch numerous evaders which are trying to escape arrest. In order to determine distributed control rules for each agent, a graph-theoretic technique is used to examine how the agents with restricted sensing capabilities interact with one another. Further, the online and real-time solution of Hamilton-Jacobi-Bellman (HJB) equations is necessary to achieve the optimal actions. By introducing the Q-learning technique's policy iteration, we are able to solve this issue. Once capture, the game is over. The simulation results are shown to demonstrate the viability of the suggested techniques.**

***Index Terms*—Adaptive dynamic programming, Q-learning, Pursuit-evasion game.**

I. Introduction

FOR SEVERAL decades, PE games have been the focus of ongoing research because they are used in many different systems, including robotic systems, autonomous drive systems, and unmanned aerial vehicles [1], [2], [3]. In particular, the PE games can depict a scenario in which one or more agents, known as pursuers, attempt to capture one or more antagonistic agents, known as evaders, while the evaders fight to escape being caught. In recent years, there has been a surge in research dedicated to pursuit-evasion (PE) games that may describe scenarios including search and rescue missions, port defense, movement camouflage and surveillance missions.

Since multi-agent systems (MASs) introduce a novel capability to complete a succession of complicated missions that cannot be achieved with a single agent/system, MASs have garnered increasing attention over the past few decades [4], [5]. Among the various multi-agent systems applications, one of the biggest problems that has an enormous effect on both the military and civilian sectors of the economy is the multiplayer pursuit-evasion game, such as guided missile and defense systems, intelligent drone control and autonomous tracking control of vehicles. The pursuit-evasion game with several large-scale pursuers and evaders has been solved in [6] using a novel actor-critic-mass-opponent strategy in a distributed manner. In [7], the study delved into the performance of players in multi-agent pursuit-evasion games on communication graphs, exploring performance indices that encompass both cooperative and antagonistic goals among the agents. A noteworthy example within this context is the construction of error dynamics for both pursuers and evaders. However, so far, few researchers have obtained the optimal control strategy and cost function of multi-agent pursuit-evasion games by numerical calculation. In this brief, an online policy iteration algorithm will be designed to approximate the optimal control strategy and cost function in real time.

The optimal control problem is solved using adaptive dynamic programming (ADP), which is regarded as an effective control method [8], [9], [10]. By resolving the coupled HJB equations, numerous analysis and synthesis findings about the best consensus designs for multi-agent systems have been obtained [11], [12]. The best course of action is determined using the model-free reinforcement learning technique called Q-learning, which draws on previous state and action observations. Q-learning algorithm is one of the most commonly used ADP algorithms for linear systems. The fact that Q-learning does not require an environment model is its greatest strength. The tracking control issues of multi-agent systems with uncertain dynamics were examined in [13]. To solve this issue, a multi-threading iterative Q-learning algorithm was created. In order to find the Pareto solutions using Q-learning algorithm, literature [14] investigated the cooperative differential game for continuous-time linear stochastic systems with markovian jumps. In this brief, we will design Q-learning algorithm to solve the numerical solution of HJB equation of pursuit-evasion game to obtain the optimal control scheme and cost function for coupled multi-agent systems. Each pursuer engages in a zero-sum game with an adversary composed of other pursuers and evaders.

The innovations are embodied in the following three aspects:

1) Different from the general multi-agent systems [15], [16], we regard neighbors and enemies as disturbances in order to transform the original pursuit-evasion game problem into a zero-sum game problem with a coupled multi-agent system.
2) Each of these three communication diagrams represents the interaction between the pursuers and the evaders, as well as the interaction between the two groups. By changing the agents' aims in relation to each graph, this approach enables us to evaluate various agent behaviors.
3) This brief is the first time to use Q-learning technology to solve the pursuit game problem of coupled multi-agent system, which relaxes the requirements of system matrices.

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The rest of this brief is organized as follows. The problem of interest for Section II is first formulated. An online model-free policy iteration algorithm is given in Section III. Section IV gives a simulation example. Finally, the conclusion is obtained in Section V.

Notation: $\otimes$ represents the Kronecker product. For a symmetric matrix $A = (a_{ij})$ is $n \times n$-dimensional, $\text{vech}(A) = [a_{11}, \ldots, a_{n1}, \ldots, a_{n1}, \ldots, a_{nn}]^{\mathrm{T}}$. $\nabla f(x) = \frac{\partial f(x)}{\partial x}$ and $\nabla^2 f(x) = \frac{\partial^2 f(x)}{\partial x^2}$.

II. Preliminaries

Define the graph as $\mathscr{G}_p = (\Pi_p, \wp_p)$, with the N pursuers being the set of nodes $\Pi_p = \{\pi_{p1}, \ldots, \pi_{pN}\}$ and a set of edges $\wp_p \subseteq \Pi_p \times \Pi_p$. Let a_{ik} represent the graph $\mathscr{G}_p$'s connectivity weights. If $(\pi_{pk}, \pi_{pi}) \in \wp_p$,$a_{ik} = 1$ and $a_{ik} = 0$; otherwise, it is equal to 0. Define the graph's in-degree matrix as $\mathcal{D}_p = \text{diag}\{d_i^p\}$ and the pursuer's in-degree matrix as $d_i^p = \sum_{k=1}^{N} a_{ik}$. $\mathcal{A}_p = [a_{ik}]$ is the weighted adjacency matrix. The graph Laplacian matrix $\mathcal{L}_p$ is defined as $\mathcal{L}_p = \mathcal{D}_p - \mathcal{A}_p$.

Similar to this, the graph $\mathscr{G}_e = (\Pi_e, \wp_e)$ depicts the interactions between the M evaders as nodes $\Pi_e = \{\pi_{e1}, \ldots, \pi_{eN}\}$. The weights of the graph are b_{jl}, where $b_{jl} = 1$ if $(\pi_{el}, \pi_{ej}) \in \wp_e$ and $b_{jl} = 0$ otherwise. The degree of the evader j is $d_j^e = \sum_{l=1}^{M} b_{jl}$. The matrices $\mathcal{D}_e = \text{diag}\{d_j^e\}$, $\mathcal{A}_e = [b_{jl}]$, and $\mathcal{L}_e = \mathcal{D}_e - \mathcal{A}_e$ are defined.

Finally, specify the bipartite graph $\mathscr{G}_{pe} = (\Pi_{pe}, \wp_e)$ that contains all of the game's participants, with the sets of N pursuers and M evaders serving as the graph's two divisions. The information flow between the agents in the various divisions is depicted in the graph $\mathscr{G}_{pe}$. In particular, edge weights c_{ij} represent the pursuer's knowledge of the evader's position. If the pursuer is able to measure the evader,$c_{ij} = 1$; otherwise, it is equal to 0. Similar to this, e_{ji} denotes the knowledge of the evader j regarding the location of the pursuer i. In the graph $\mathscr{G}_{pe}$, the in-degree of the pursuer i is defined as $d_i^{pe} = \sum_{j=1}^{M} c_{ij}$, and the in-degree of the evader j is $d_j^{ep} = \sum_{i=1}^{N} e_{ji}$. Define the pursuer-partition matrices $\mathcal{D}_{pe} = \text{diag}\{d_i^{pe}\}$ and $\mathcal{A}_{pe} = [c_{ij}]$, as well as the evader-partition matrices $\mathcal{D}_{ep} = \text{diag}\{d_j^{ep}\}$ and $\mathcal{A}_{ep} = [e_{ji}]$.

Consider a group of N dynamical agents with linear dynamics which are the pursuers as follows:

$$\dot{x}_i = Ax_i + B\mu_i, \quad i = 1, \ldots, N \tag{1}$$

where $x_i \in \mathbb{R}^n$ is the position of pursuer i. The set of M dynamic agents is viewed as a team of evaders with dynamics

$$\dot{y}_j = Ay_j + B\nu_j, \quad j = 1, \ldots, M \tag{2}$$

where $y_j \in \mathbb{R}^n$ is the position.

The pursuers need to protect the target and intercept the pursuers while maintaining the aggregation state. Therefore, the weighted distance that the pursuer i need to optimize can be expressed as:

$$\varepsilon_i = \sum_{k=1}^{N} a_{ik}(x_i - x_k) + \sum_{j=1}^{M} c_{ij}\big(x_i - y_j\big). \tag{3}$$

The dynamics of the local error variable ε_i for pursuer i is given by

$$\dot{\varepsilon}_i = A\varepsilon_i + \big(d_i^p + d_i^{pe}\big)B\mu_i - \big(d_i^p + d_i^{pe}\big)B\bar{\mu}_{-i} \tag{4}$$

where

$$\bar{\mu}_{-i} = \frac{1}{d_i^p + d_i^{pe}}\left(\sum_{k=1}^{N} a_{ik}\mu_k + \sum_{j=1}^{M} c_{ij}\nu_j\right).$$

The local position error for evader j is similarly defined as

$$\epsilon_j = \sum_{l=1}^{M} b_{jl}\big(y_j - y_l\big) - \sum_{i=1}^{N} e_{ji}\big(y_j - x_i\big). \tag{5}$$

And the local error dynamics change as

$$\dot{\epsilon}_j = A\epsilon_j + \left(d_j^e - d_j^{ep}\right)B\nu_j - \left(d_j^e - d_j^{ep}\right)B\bar{\nu}_{-j} \tag{6}$$

where

$$\bar{\nu}_{-j} = \frac{1}{d_j^e - d_j^{ep}}\left(\sum_{l=1}^{M} b_{jl}\nu_l \;\; - \sum_{i=1}^{N} e_{ji}\mu_i\right).$$

The performance metrics that we can define for the pursuer i are

$$\begin{aligned}\mathscr{V}_{pi} &= \int_t^{\infty} r_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i})d\tau \\ &= \int_t^{\infty} \big[\varepsilon_i^{\mathrm{T}} Q_{pi}\varepsilon_i + \mu_i^{\mathrm{T}} R_p\mu_i - \bar{\mu}_{-i}^{\mathrm{T}} R_{-i}\bar{\mu}_{-i}\big]d\tau \end{aligned} \tag{7}$$

where Q_{pi}, R_p and R_{-i} are positive definite matrices. Similarly, for the evader j, the design the performance index is

$$\begin{aligned}\mathscr{V}_{ej} &= \int_t^{\infty} r_{ej}\big(\epsilon_j, \nu_j, \bar{\nu}_{-j}\big)d\tau \\ &= \int_t^{\infty} \Big[\epsilon_j^{\mathrm{T}} Q_{ej}\epsilon_j + \nu_j^{\mathrm{T}} R_e\nu_j - \bar{\nu}_{-j}^{\mathrm{T}} R_{-j}\bar{\nu}_{-j}\Big]d\tau \end{aligned} \tag{8}$$

where Q_{ej}, R_e and R_{-j} are positive definite matrices.

The performance indexes can both be recast by positive-definite quadratic functions in accordance with the optimum control theory [17] where $\mathcal{P}_{pi}$ and $\mathcal{P}_{ej}$ are positive-definite matrices, and $\mathscr{V}_{pi} = \varepsilon_i^{\mathrm{T}}\mathcal{P}_{pi}\varepsilon_i$ and $\mathscr{V}_{ej} = \epsilon_j^{\mathrm{T}}\mathcal{P}_{ej}\epsilon_j$, respectively.

Given neighbor strategies μ_k and ν_j, the ideal control strategy of pursuer i can be obtained by solving the ith Hamiltonian function defined as

$$\begin{aligned}H_{pi} = \nabla\mathscr{V}_{pi}^{\mathrm{T}}&\Big(A\varepsilon_i + \big(d_i^p + d_i^{pe}\big)B\mu_i - \big(d_i^p + d_i^{pe}\big)B\bar{\mu}_{-i}\Big) \\ &+ \varepsilon_i^{\mathrm{T}} Q_{pi}\varepsilon_i + \mu_i^{\mathrm{T}} R_p\mu_i - \bar{\mu}_{-i}^{\mathrm{T}} R_{-i}\bar{\mu}_{-i}\end{aligned} \tag{9}$$

and it is found to be

$$\mu_i^* = -\big(d_i^p + d_i^{pe}\big)R_p^{-1}B^{\mathrm{T}}\mathcal{P}_{pi}\varepsilon_i, \tag{10}$$

$$\bar{\mu}_i^* = -\big(d_i^p + d_i^{pe}\big)R_{-i}^{-1}B^{\mathrm{T}}\mathcal{P}_{pi}\varepsilon_i. \tag{11}$$

Define the Hamiltonian function for evader j as follows using the same method:

$$\begin{aligned}H_{ej} = \nabla\mathscr{V}_{ej}^{\mathrm{T}}&\Big(A\epsilon_j + \left(d_j^e - d_j^{ep}\right)B\nu_j - \left(d_j^e - d_j^{ep}\right)B\bar{\nu}_{-j}\Big) \\ &+ \epsilon_j^{\mathrm{T}} Q_{ej}\epsilon_j + \nu_j^{\mathrm{T}} R_e\nu_j - \bar{\nu}_{-j}^{\mathrm{T}} R_{-j}\bar{\nu}_{-j}\end{aligned} \tag{12}$$

and the optimal policy for evader j is

$$\nu_j^* = -\left(d_j^e - d_j^{ep}\right)R_e^{-1}B^{\mathrm{T}}\mathcal{P}_{ej}\epsilon_j, \tag{13}$$

$$\bar{\nu}_{-j}^* = -\left(d_j^e - d_j^{ep}\right)R_{-j}^{-1}B^{\mathrm{T}}\mathcal{P}_{ej}\epsilon_j. \tag{14}$$

III. $\mathcal{Q}$-Learning Pursuer-Evader Game Algorithm

To express the pursuer $\mathcal{Q}$-function, the optimal value function $\mathscr{V}_{pi}^{*} = \min_{\mu_i}\max_{\mu_{-i}}\mathscr{V}_{pi}$ must be parameterized as a function of the state ε_i and the control μ_i. The following action-dependent value $\mathcal{Q}_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i})$ can be written:

$$\begin{aligned}\mathcal{Q}_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i}) &= \mathscr{V}_{pi}^{*} + H_{pi}\\ &= \varepsilon_i^{\mathrm{T}}\mathcal{P}_{pi}\big[A\varepsilon_i + \big(d_i^{p} + d_i^{pe}\big)B\mu_i - \big(d_i^{p} + d_i^{pe}\big)B\bar{\mu}_{-i}\big]\\ &\quad + \big[A\varepsilon_i + \big(d_i^{p} + d_i^{pe}\big)B\mu_i - \big(d_i^{p} + d_i^{pe}\big)B\bar{\mu}_{-i}\big]^{\mathrm{T}}\mathcal{P}_{pi}\varepsilon_i\\ &\quad + \varepsilon_i^{\mathrm{T}}Q_{pi}\varepsilon_i + \mu_i^{\mathrm{T}}R_p\mu_i - \bar{\mu}_{-i}^{\mathrm{T}}R_{-i}\bar{\mu}_{-i} + \varepsilon_i^{\mathrm{T}}\mathcal{P}_{pi}\varepsilon_i. \end{aligned} \tag{15}$$

Similarly, $\mathcal{Q}_{ej}(\epsilon_i, \nu_j, \bar{\nu}_{-j})$ is given by

$$\begin{aligned}\mathcal{Q}_{ej}\big(\epsilon_i, \nu_j, \bar{\nu}_{-j}\big) &= \mathscr{V}_{ej}^{*} + H_{ej}\\ &= \epsilon_j^{\mathrm{T}}\mathcal{P}_{ej}\Big[A\epsilon_j + \Big(d_j^{e} - d_j^{ep}\Big)B\nu_j - \Big(d_j^{e} - d_j^{ep}\Big)B\bar{\nu}_{-j}\Big]\\ &\quad + \Big[A\epsilon_j + \Big(d_j^{e} - d_j^{ep}\Big)B\nu_j - \Big(d_j^{e} - d_j^{ep}\Big)B\bar{\nu}_{-j}\Big]^{\mathrm{T}}\mathcal{P}_{ej}\varepsilon_i\\ &\quad + \epsilon_j^{\mathrm{T}}Q_{ej}\epsilon_j + \nu_j^{\mathrm{T}}R_e\nu_j - \bar{\nu}_{-j}^{\mathrm{T}}R_{-j}\bar{\nu}_{-j} + \epsilon_j^{\mathrm{T}}\mathcal{P}_{ej}\epsilon_j. \end{aligned} \tag{16}$$

The action-dependent function (15) can be rewritten in the state-action form using a compact quadratic as

$$\begin{aligned}\mathcal{Q}_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i}) &= \bar{\mathcal{X}}_{pi}^{\mathrm{T}}\bar{\mathcal{Q}}_{pi}\bar{\mathcal{X}}_{pi} \qquad (17)\\ &= \begin{bmatrix}\varepsilon_i\\ \mu_i\\ \bar{\mu}_{-i}\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\mathcal{Q}_{pi}^{\varepsilon_i\varepsilon_i} & \mathcal{Q}_{pi}^{\varepsilon_i\mu_i} & \mathcal{Q}_{pi}^{\varepsilon_i\bar{\mu}_{-i}}\\ \mathcal{Q}_{pi}^{\mu_i\varepsilon_i} & \mathcal{Q}_{pi}^{\mu_i\mu_i} & \mathcal{Q}_{pi}^{\mu_i\bar{\mu}_{-i}}\\ \mathcal{Q}_{pi}^{\bar{\mu}_{-i}\varepsilon_i} & \mathcal{Q}_{pi}^{\bar{\mu}_{-i}\mu_i} & \mathcal{Q}_{pi}^{\bar{\mu}_{-i}\bar{\mu}_{-i}}\end{bmatrix}\begin{bmatrix}\varepsilon_i\\ \mu_i\\ \bar{\mu}_{-i}\end{bmatrix}\end{aligned}$$

where

$$\begin{aligned}\mathcal{Q}_{pi}^{\varepsilon_i\varepsilon_i} &= \mathcal{P}_{pi}A + A^{\mathrm{T}}\mathcal{P}_{pi} + Q_{pi} + \mathcal{P}_{pi},\\ \mathcal{Q}_{pi}^{\varepsilon_i\mu_i} &= \mathcal{P}_{pi}\big(d_i^{p} + d_i^{pe}\big)B,\ \mathcal{Q}_{pi}^{\varepsilon_i\bar{\mu}_{-i}} = -\mathcal{P}_{pi}\big(d_i^{p} + d_i^{pe}\big)B,\\ \mathcal{Q}_{pi}^{\mu_i\varepsilon_i} &= B^{\mathrm{T}}\big(d_i^{p} + d_i^{pe}\big)\mathcal{P}_{pi},\ \mathcal{Q}_{pi}^{\mu_i\mu_i} = R_p,\\ \mathcal{Q}_{pi}^{\bar{\mu}_{-i}\varepsilon_i} &= -B^{\mathrm{T}}\big(d_i^{p} + d_i^{pe}\big)\mathcal{P}_{pi},\ \mathcal{Q}_{pi}^{\bar{\mu}_{-i}\bar{\mu}_{-i}} = -R_{-i},\end{aligned}$$

and

$$\begin{aligned}\mathcal{Q}_{ej}\big(\epsilon_j, \nu_j, \bar{\nu}_{-j}\big) &= \bar{\mathcal{X}}_{ej}^{\mathrm{T}}\bar{\mathcal{Q}}_{ej}\bar{\mathcal{X}}_{ej} \qquad (18)\\ &= \begin{bmatrix}\epsilon_j\\ \nu_j\\ \bar{\nu}_{-j}\end{bmatrix}^{\mathrm{T}}\begin{bmatrix}\mathcal{Q}_{ej}^{\epsilon_j\epsilon_j} & \mathcal{Q}_{ej}^{\epsilon_j\nu_j} & \mathcal{Q}_{ej}^{\epsilon_j\bar{\nu}_{-j}}\\ \mathcal{Q}_{ej}^{\nu_j\epsilon_j} & \mathcal{Q}_{ej}^{\nu_j\nu_j} & \mathcal{Q}_{ej}^{\nu_j\bar{\nu}_{-j}}\\ \mathcal{Q}_{ej}^{\bar{\nu}_{-j}\epsilon_j} & \mathcal{Q}_{ej}^{\bar{\nu}_{-j}\nu_j} & \mathcal{Q}_{ej}^{\bar{\nu}_{-j}\bar{\nu}_{-j}}\end{bmatrix}\begin{bmatrix}\epsilon_j\\ \nu_j\\ \bar{\nu}_{-j}\end{bmatrix}\end{aligned}$$

where

$$\begin{aligned}\mathcal{Q}_{ej}^{\epsilon_j\epsilon_j} &= \mathcal{P}_{ej}A + A^{\mathrm{T}}\mathcal{P}_{ej} + Q_{ej} + \mathcal{P}_{ej},\\ \mathcal{Q}_{ej}^{\epsilon_j\nu_j} &= \mathcal{P}_{ej}\Big(d_j^{e} - d_j^{ep}\Big)B,\ \mathcal{Q}_{ej}^{\epsilon_j\bar{\nu}_{-j}} = -\mathcal{P}_{ej}\Big(d_j^{e} - d_j^{ep}\Big)B,\\ \mathcal{Q}_{ej}^{\nu_j\epsilon_j} &= B^{\mathrm{T}}\Big(d_j^{e} - d_j^{ep}\Big)\mathcal{P}_{ej},\ \mathcal{Q}_{ej}^{\nu_j\nu_j} = R_e,\\ \mathcal{Q}_{ej}^{\bar{\nu}_{-j}\epsilon_j} &= -B^{\mathrm{T}}\Big(d_j^{e} + d_j^{ep}\Big)\mathcal{P}_{ej},\ \mathcal{Q}_{ej}^{\bar{\nu}_{-j}\bar{\nu}_{-j}} = -R_{-j}.\end{aligned}$$

Lemma 1: Given that the Q-functions are equivalent as in (17)–(18) and that $\mathscr{V}_{pi}^{*}$ and $\mathscr{V}_{ej}^{*}$ are the ideal value functions and $\mathcal{P}_{pi}$ and $\mathcal{P}_{ej}$ are the answers respectively to the Riccati equations

$$\mathcal{Q}_{pi}^{*}\big(\varepsilon_i, \mu_i^{*}, \bar{\mu}_{-i}^{*}\big) = \mathscr{V}_{pi}^{*} = \min_{\mu_i}\max_{\bar{\mu}_{-i}}\mathscr{V}_{pi},$$

$$\mathcal{Q}_{ej}^{*}\Big(\epsilon_j, \nu_j^{*}, \bar{\nu}_{-j}^{*}\Big) = \mathscr{V}_{ej}^{*} = \min_{\nu_j}\max_{\bar{\nu}_{-j}}\mathscr{V}_{ej}.$$

We solve $\frac{\partial\mathcal{Q}_{pi}(\varepsilon_i,\mu_i,\bar{\mu}_{-i})}{\partial\mu_i}$ and $\frac{\partial\mathcal{Q}_{pi}(\varepsilon_i,\mu_i,\bar{\mu}_{-i})}{\partial\bar{\mu}_{-i}}$ for the control and disturbance policy to obtain

$$\begin{aligned}\mu_i^{*} &= \arg\min_{\mu_i}\mathcal{Q}_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i})\\ &= -\Big(\mathcal{Q}_{pi}^{\mu_i\mu_i}\Big)^{-1}\mathcal{Q}_{pi}^{\mu_i\varepsilon_i}\varepsilon_i\\ &= \Big(\Gamma_{pi}^{\mu_i}\Big)^{\mathrm{T}}\varepsilon_i, \qquad (19)\\ \bar{\mu}_{-i}^{*} &= \arg\max_{\bar{\mu}_{-i}}\mathcal{Q}_{pi}(\varepsilon_i, \mu_i, \bar{\mu}_{-i})\\ &= -\Big(\mathcal{Q}_{pi}^{\bar{\mu}_{-i}\bar{\mu}_{-i}}\Big)^{-1}\mathcal{Q}_{pi}^{\bar{\mu}_{-i}\varepsilon_i}\varepsilon_i \qquad (20)\end{aligned}$$

where $\Gamma_{pi}^{\mu_i}$ is the ideal weight for optimal feedback for the pursuer i.

For evader j, the same procedure can be used to prove that

$$\begin{aligned}\nu_j^{*} &= \arg\min_{\nu_j}\mathcal{Q}_{ej}\big(\epsilon_j, \nu_j, \bar{\nu}_{-j}\big)\\ &= -\Big(\mathcal{Q}_{ej}^{\nu_j\nu_j}\Big)^{-1}\mathcal{Q}_{ej}^{\nu_j\epsilon_i}\epsilon_j\\ &= \Big(\Gamma_{ej}^{\nu_j}\Big)^{\mathrm{T}}\epsilon_j, \qquad (21)\\ \bar{\nu}_{-j}^{*} &= \arg\max_{\bar{\nu}_{-j}}\mathcal{Q}_{ej}\big(\epsilon_j, \nu_j, \bar{\nu}_{-j}\big)\\ &= -\Big(\mathcal{Q}_{ej}^{\bar{\nu}_{-j}\bar{\nu}_{-j}}\Big)^{-1}\mathcal{Q}_{ej}^{\bar{\nu}_{-j}\epsilon_j}\epsilon_j \qquad (22)\end{aligned}$$

where $\Gamma_{ej}^{\nu_j}$ is the ideal weights for optimal feedback for the evader j.

Using the elements in $\bar{\mathcal{X}}_{pi}\otimes\bar{\mathcal{X}}_{pi}$ and $\bar{\mathcal{X}}_{ej}\otimes\bar{\mathcal{X}}_{ej}$ as bases, they are rewrite in a compact form as

$$\begin{aligned}\mathcal{Q}_{pi}^{*}\big(\varepsilon_i, \mu_i^{*}, \bar{\mu}_{-i}^{*}\big) &= \mathrm{vech}\big(\bar{\mathcal{Q}}_{pi}\big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{pi}\otimes\bar{\mathcal{X}}_{pi}\big)\\ &= \Big(\Gamma_{pi}^{c}\Big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{pi}\otimes\bar{\mathcal{X}}_{pi}\big), \qquad (23)\\ \mathcal{Q}_{ej}^{*}\Big(\epsilon_i, \nu_j^{*}, \bar{\nu}_{-j}^{*}\Big) &= \mathrm{vech}\big(\bar{\mathcal{Q}}_{ej}\big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{ej}\otimes\bar{\mathcal{X}}_{ej}\big)\\ &= \Big(\Gamma_{ej}^{c}\Big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{ej}\otimes\bar{\mathcal{X}}_{ej}\big). \qquad (24)\end{aligned}$$

The following weight estimates for the critic approximator must be taken into consideration because the ideal weights for computing $\mathcal{Q}_{pi}^{*}$ and $\mathcal{Q}_{ej}^{*}$ are unknown

$$\hat{\mathcal{Q}}_{pi}^{*}\big(\varepsilon_i, \mu_i^{*}, \bar{\mu}_{-i}^{*}\big) = \Big(\hat{\Gamma}_{pi}^{c}\Big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{pi}\otimes\bar{\mathcal{X}}_{pi}\big), \tag{25}$$

$$\hat{\mathcal{Q}}_{ej}^{*}\Big(\epsilon_i, \nu_j^{*}, \bar{\nu}_{-j}^{*}\Big) = \Big(\hat{\Gamma}_{ej}^{c}\Big)^{\mathrm{T}}\big(\bar{\mathcal{X}}_{ej}\otimes\bar{\mathcal{X}}_{ej}\big) \tag{26}$$

where $\hat{\Gamma}_{pi}^{c}$ and $\hat{\Gamma}_{ej}^{c}$ are the estimated weights.

The actual control strategies are

$$\hat{\mu}_i^{*} = \Big(\hat{\Gamma}_{pi}^{\mu_i}\Big)^{\mathrm{T}}\varepsilon_i,\ \ \hat{\bar{\mu}}_{-i}^{*} = -\Big(\hat{\mathcal{Q}}_{pi}^{\bar{\mu}_{-i}\bar{\mu}_{-i}}\Big)^{-1}\hat{\mathcal{Q}}_{pi}^{\bar{\mu}_{-i}\varepsilon_i}\varepsilon_i, \tag{27}$$

$$\hat{\nu}_j^{*} = \Big(\hat{\Gamma}_{ej}^{\nu_j}\Big)^{\mathrm{T}}\epsilon_j,\ \ \hat{\bar{\nu}}_{-j}^{*} = -\Big(\hat{\mathcal{Q}}_{ej}^{\bar{\nu}_{-j}\bar{\nu}_{-j}}\Big)^{-1}\hat{\mathcal{Q}}_{ej}^{\bar{\nu}_{-j}\epsilon_j}\epsilon_j \tag{28}$$

where $\hat{\Gamma}_{pi}^{\mu_i}$ and $\hat{\Gamma}_{ej}^{\nu_j}$ and are the weight estimates, the values of $\hat{\mathcal{Q}}_{pi}^{\bar{\mu}_{-i}\bar{\mu}_{-i}}$, $\hat{\mathcal{Q}}_{pi}^{\bar{\mu}_{-i}\varepsilon_i}$ and $\hat{\mathcal{Q}}_{ej}^{\bar{\nu}_{-j}\bar{\nu}_{-j}}$, $\hat{\mathcal{Q}}_{ej}^{\bar{\nu}_{-j}\epsilon_j}$ are going to be extracted from the vector $\hat{\Gamma}_{pi}^{c}$ and $\hat{\Gamma}_{ej}^{c}$, respectively.

According to the integral reinforcement learning approach, $T > 0$ is a small integral time interval, consequently, using Lemma 1, we have

$$\begin{aligned}&\mathscr{Q}^*_{pi}\big(\varepsilon_i(t), \mu^*_i(t), \bar{\mu}^*_{-i}(t)\big)\\&\quad= \int_t^{t+T} r_{pi}\big(\varepsilon_i, \mu^*_i, \bar{\mu}^*_{-i}\big)d\tau\\&\qquad+ \mathscr{Q}^*_{pi}\big(\varepsilon_i(t+T), \mu^*_i(t+T), \bar{\mu}^*_{-i}(t+T)\big), \quad (29)\end{aligned}$$

$$\begin{aligned}&\mathscr{Q}^*_{ej}\Big(\epsilon_j(t), \nu^*_j(t), \bar{\nu}^*_{-j}(t)\Big)\\&\quad= \int_t^{t+T} r_{ej}\Big(\epsilon_j, \nu^*_j, \bar{\nu}^*_{-j}\Big)d\tau\\&\qquad+ \mathscr{Q}^*_{ej}\Big(\epsilon_j(t+T), \nu^*_j(t+T), \bar{\nu}^*_{-j}(t+T)\Big). \quad (30)\end{aligned}$$

Based on the current estimate of the action-dependent function that we want to drive to zero, we will now define the error as

$$\begin{aligned}e^c_{pi} &= -\hat{\mathscr{Q}}^*_{pi}\big(\varepsilon_i(t), \mu^*_i(t), \bar{\mu}^*_{-i}(t)\big) + \int_t^{t+T} r_{pi}\big(\varepsilon_i, \mu^*_i, \bar{\mu}^*_{-i}\big)d\tau\\&\quad+ \hat{\mathscr{Q}}^*_{pi}\big(\varepsilon_i(t+T), \mu^*_i(t+T), \bar{\mu}^*_{-i}(t+T)\big)\\&= \Big(\hat{\Gamma}^c_{pi}\Big)^{\mathrm{T}}\sigma_{pi} + \int_t^{t+T} r_{pi}\big(\varepsilon_i, \mu^*_i, \bar{\mu}^*_{-i}\big)d\tau, \quad (31)\end{aligned}$$

$$\begin{aligned}e^c_{ej} &= -\hat{\mathscr{Q}}^*_{ej}\Big(\epsilon_j(t), \nu^*_j(t), \bar{\nu}^*_{-j}(t)\Big) + \int_t^{t+T} r_{ej}\Big(\epsilon_j, \nu^*_j, \bar{\nu}^*_{-j}\Big)d\tau\\&\quad+ \hat{\mathscr{Q}}^*_{ej}\Big(\epsilon_j(t+T), \nu^*_j(t+T), \bar{\nu}^*_{-j}(t+T)\Big)\\&= \Big(\hat{\Gamma}^c_{ej}\Big)^{\mathrm{T}}\sigma_{ej} + \int_t^{t+T} r_{ej}\Big(\epsilon_j, \nu^*_j, \bar{\nu}^*_{-j}\Big)d\tau, \quad (32)\end{aligned}$$

where $\sigma_{pi} = \bar{\mathcal{X}}_{pi}(t+T) \otimes \bar{\mathcal{X}}_{pi}(t+T) - \bar{\mathcal{X}}_{pi}(t) \otimes \bar{\mathcal{X}}_{pi}(t)$ and $\sigma_{ej} = \bar{\mathcal{X}}_{ej}(t+T) \otimes \bar{\mathcal{X}}_{ej}(t+T) - \bar{\mathcal{X}}_{ej}(t) \otimes \bar{\mathcal{X}}_{ej}(t)$.

We choose the loss function as

$$E^c_{pi} = \frac{1}{2}\|e^c_{pi}\|^2, \quad E^c_{ej} = \frac{1}{2}\|e^c_{ej}\|^2.$$

We use gradient descent and normalization to get the dynamics for estimating the critic weights as

$$\dot{\hat{\Gamma}}^c_{pi} = -\alpha^c_{pi}\frac{\partial E^c_{pi}}{\partial \hat{\Gamma}^c_{pi}} = -\alpha^c_{pi}\frac{\sigma_{pi}}{1+\sigma^{\mathrm{T}}_{pi}\sigma_{pi}}\Big(e^c_{pi}\Big)^{\mathrm{T}}, \quad (33)$$

$$\dot{\hat{\Gamma}}^c_{ej} = -\alpha^c_{ej}\frac{\partial E^c_{ej}}{\partial \hat{\Gamma}^c_{ej}} = -\alpha^c_{ej}\frac{\sigma_{ej}}{1+\sigma^{\mathrm{T}}_{ej}\sigma_{ej}}\Big(e^c_{ej}\Big)^{\mathrm{T}} \quad (34)$$

where the learning steps are $\alpha^c_{pi} > 0$ and $\alpha^c_{ej} > 0$.

In the actor model, we choose the error functions as follows:

$$E^{\mu_i}_{pi} = \frac{1}{2}\|e^{\mu_i}_{pi}\|^2, \quad E^{\nu_j}_{ej} = \frac{1}{2}\|e^{\nu_j}_{ej}\|^2$$

where $e^{\mu_i}_{pi} = (\hat{\Gamma}^{\mu_i}_{pi})^{\mathrm{T}}\varepsilon_i + (\hat{\mathscr{Q}}^{\mu_i\mu_i}_{pi})^{-1}\hat{\mathscr{Q}}^{\mu_i\varepsilon_i}_{pi}\varepsilon_i$ and $e^{\nu_j}_{ej} = (\hat{\Gamma}^{\nu_j}_{ej})^{\mathrm{T}}\epsilon_j + (\hat{\mathscr{Q}}^{\nu_j\nu_j}_{ej})^{-1}\hat{\mathscr{Q}}^{\nu_j\epsilon_j}_{ej}\epsilon_j$.

Using the gradient descent, it produces the desired result to guarantee $e^{\mu_i}_{pi} \to 0$ and $e^{\nu_j}_{ej} \to 0$

$$\dot{\hat{\Gamma}}^{\mu_i}_{pi} = -\alpha^{\mu_i}_{pi}\frac{\partial E^{\mu_i}_{pi}}{\partial \hat{\Gamma}^{\mu_i}_{pi}} = -\alpha^{\mu_i}_{pi}\varepsilon_i\Big(e^{\mu_i}_{pi}\Big)^{\mathrm{T}}, \quad (35)$$

$$\dot{\hat{\Gamma}}^{\nu_j}_{ej} = -\alpha^{\nu_j}_{ej}\frac{\partial E^{\nu_j}_{ej}}{\partial \hat{\Gamma}^{\nu_j}_{ej}} = -\alpha^{\nu_j}_{ej}\epsilon_j\Big(e^{\nu_j}_{ej}\Big)^{\mathrm{T}} \quad (36)$$

where the learning steps are $\alpha^{\mu_i}_{pi} > 0$ and $\alpha^{\nu_j}_{ej} > 0$.

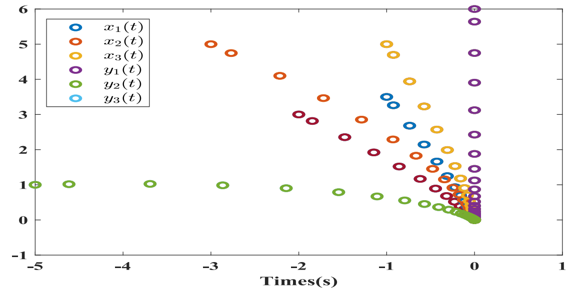

Fig. 1. Trajectories of the pursuers and evaders when attackers win.

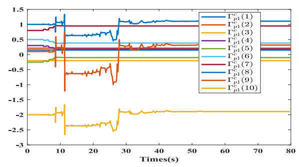

Fig. 2. The evolution process of critic NNs weights for pursuer 1.

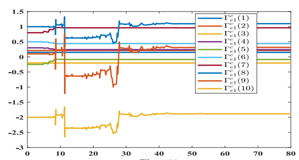

Fig. 3. The evolution process of critic NNs weights for evader 1.

IV. Example

The following section discusses numerical simulations for asymptotic capture.

Consider an pursuer-evader game with three pursuers and three evaders in $\mathbb{R}^2$ which has single-integrator dynamics, i.e., $A = -I_2$ and $B = [0; 1]$ in (1) and (2), connected in a communication network such that

$$\mathcal{L}_p = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}, \mathcal{L}_e = \begin{bmatrix} 2 & -1 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix},$$
$$\mathcal{A}_{pe} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \mathcal{L}_e = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

The NNs basis functions for this case are established as

$$\begin{aligned}\bar{\mathcal{X}}_{pi} \otimes \bar{\mathcal{X}}_{pi} &= [\varepsilon^2_i(1), 2\varepsilon_i(1)\varepsilon_i(2), \varepsilon^2_i(2), 2\varepsilon_i(1)\mu^*_i, 2\varepsilon_i(2)\mu^*_i,\\&\quad 2\varepsilon_i(1)\bar{\mu}^*_{-i}, 2\varepsilon_i(2)\bar{\mu}^*_{-i}, \mu^{*2}_i, 2\mu^*_i\bar{\mu}^*_{-i}, \bar{\mu}^{*2}_{-i}]^{\mathrm{T}},\\ \bar{\mathcal{X}}_{ej} \otimes \bar{\mathcal{X}}_{ej} &= [\epsilon^2_j(1), 2\epsilon_j(1)\epsilon_j(2), \epsilon^2_j(2), 2\epsilon_j(1)\nu^*_j, 2\epsilon_j(2)\nu^*_j,\\&\quad 2\epsilon_j(1)\bar{\nu}^*_{-j}, 2\epsilon_j(2)\bar{\nu}^*_{-j}, \nu^{*2}_j, 2\nu^*_j\bar{\nu}^*_{-j}, \bar{\nu}^{*2}_{-j}]^{\mathrm{T}}.\end{aligned}$$

The integral time interval is $T = 2s$ and the adaptive learning gains are $\alpha^c_{pi} = \alpha^c_{ej} = 1.1$ and $\alpha^{\mu_i}_{pi} = \alpha^{\nu_j}_{ej} = 0.1$. Some parameters are given as $Q_{pi} = 10I$, $Q_{ej} = 10I$, $R_p = 1$, $R_e = 1$, $R_{-i} = 1$, $R_{-j} = 1$.

Fig. 1 displays the simulation's outcome and demonstrates that capture occurs, supporting the results of the theory. The Q-learning technique eliminates the requirement for system matrices. Figs. 2-9 indicate that the weights have converged. The optimal control strategy can be obtained by multiplying with the activation function.

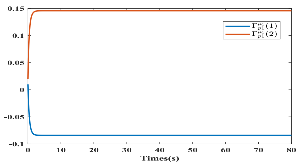

Fig. 4. The evolution process of actor NNs weights for pursuer 1.

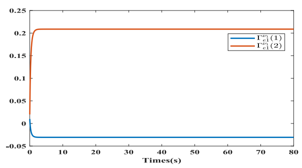

Fig. 5. The evolution process of actor NNs weights for evader 1.

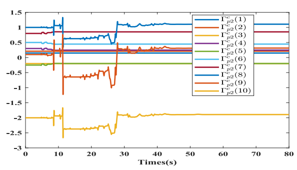

Fig. 6. The evolution process of critic NNs weights for pursuer 2.

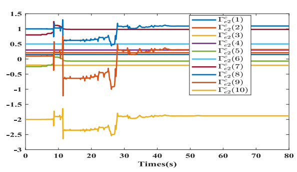

Fig. 7. The evolution process of critic NNs weights for evader 2.

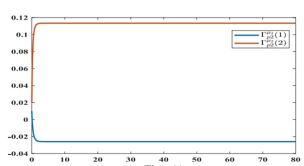

Fig. 8. The evolution process of actor NNs weights for pursuer 2.

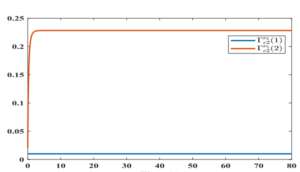

Fig. 9. The evolution process of actor NNs weights for evader 2.

V. Conclusion

In this brief, Q-learning method has been used to study the pursuit game problem of coupling multi-agent system. Firstly, when the pursuer and evader sides remained dispersed, a coupled multi-agent system has been established to transform the pursuer-evader game problem of multiple pursuers and evaders into multiple groups of two-player zero-sum differential games, and the optimal strategies of the attacking and defending sides are solved. Then, an online Q-learning algorithm has been used to obtain the numerical solutions of the HJB equations without using the system matrices. Finally, numerical simulation has been verified the effectiveness of the proposed method. In the future research, the heterogeneous nonlinear pursuer-evader game model and the switching-like event-triggered control should be considered. Nonlinearity is more realistic, but it will bring difficulties to our research.

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